

On Equivalence of Norms

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Definition 0.1 We say that two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on a vector space V are equivalent if there exist two positive constants c_1, c_2 such that

$$c_1 \|v\|_1 \leq \|v\|_2 \leq c_2 \|v\|_1 \quad \forall v \in V.$$

Remark 0.2 Notice that if the norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on the vector space V are equivalent, then the normed spaces $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ have the same converging sequences.

Theorem 0.3 *In a finite dimensional vector space, all the norms are equivalent.*

Proof: Let V be a finite dimensional vector space over the field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. Let n be the dimension of V and let $(e_1, e_2, \dots, e_n)$ be a basis of V . So for any $x \in V$, we have that $x = \sum_{k=1}^n x_k e_k$. Set

$$\|x\| = \sum_{k=1}^n |x_k|. \text{ We know that } \|\cdot\| \text{ is a norm on } V.$$

Now let N be another norm on N . We want to prove that N and $\|\cdot\|$ are equivalent. It is clear that

$$N(x) = N\left(\sum_{k=1}^n x_k e_k\right) \leq \sum_{k=1}^n |x_k| N(e_k) \leq \left[\max_{1 \leq k \leq n} N(e_k)\right] \sum_{k=1}^n |x_k| = M \|x\| \text{ where } M = \max_{1 \leq k \leq n} N(e_k).$$

Now we want to prove that there exists $\alpha > 0$ such that

$$N(x) \geq \alpha \|x\| \quad \forall x \in V.$$

which is equivalent to prove that

$$N\left(\sum_{k=1}^n x_k e_k\right) \geq \alpha \quad \forall (x_1, \dots, x_n) \text{ such that } \sum_{k=1}^n |x_k| = 1.$$

Assume by contradiction that this is false. Hence, for any $m \in \mathbb{N}^+$,

$$\exists x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)}) \in \mathbb{R}^n \text{ such that } N\left(\sum_{k=1}^n x_k^{(m)} e_k\right) < \frac{1}{m} \quad \text{and} \quad \sum_{k=1}^n |x_k^{(m)}| = 1.$$

So, $(x_1^m, \dots, x_n^m)_m$ is a bounded sequence in $\mathbb{R}^n$ and hence, there exists a subsequence $(x^{(m_j)})$ which converges to some $(x_1, \dots, x_n) \in \mathbb{R}^n$.

(Here, we applied componentwise Bolzano-Weierstrass' theorem: starting from the boundedness of (x_1^m) we got a subsequence $(x_1^{m_k})$ which converges to x_1 , then from the boundedness of $(x_2^{m_k})$ we got a further subsequence which converges to x_2 and so on...)

$$\text{Set } x := \sum_{k=1}^n x_k e_k.$$

Therefore,

$$\left\{ \begin{array}{l} \|\|x^{m_j}\| - \|x\| \| \leq \|x^{m_j} - x\| \rightarrow 0 \text{ and} \\ |N(x^{m_j}) - N(x)| \leq N(x^{m_j} - x) \leq M \|x^{m_j} - x\| \rightarrow 0. \end{array} \right. \quad (*)$$

So we get from

$$\left. \begin{array}{l} N(x^{(m_j)}) < \frac{1}{m_j} \\ \|x^{(m_j)}\| = 1 \quad \forall j \in \mathbb{N} \setminus \{0\} \end{array} \right\}$$

and from $(*)$ (passing to the limit as $j \rightarrow \infty$) that

$$N(x) = 0 \quad \text{and} \quad \|x\| = 1 \quad \Rightarrow \quad x = 0_V \quad \text{and} \quad \|x\| = 1.$$

which is a contradiction. ■

Remark 0.4 The statement in Theorem 0.3 does not always hold on infinite dimensional vector spaces. **Give an example of norms which are not equivalent on infinite dimensional spaces!**

Exercise 0.5 Let $(V, +, \cdot)$ be a finite dimensional vector space over $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Show that all linear mappings from V to $\mathbb{K}$ are continuous.

Exercise 0.6 Let $p \in [1, \infty)$. Let $\|\cdot\|_p$ and $\|\cdot\|_\infty$ be the norms defined on $\mathbb{R}^d$ by

$$\|x\|_p := \left(\sum_{i=1}^d |x_i|^p \right)^{1/p} \quad \text{and} \quad \|x\|_\infty := \sup_{1 \leq i \leq d} |x_i| \quad \forall x = (x_1, \dots, x_d) \in \mathbb{R}^d.$$

Prove that

$$\lim_{p \rightarrow \infty} \|v\|_p = \|v\|_\infty \quad \forall v \in V.$$

[Hint: Use the equivalence of these norms.]

Exercise 0.7 Let $(V, +, \cdot)$ be a vector space on the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$ and let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V with d_1 and d_2 denoting the corresponding distances induced. Prove that the following statements are equivalent:

- (a) the norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent;
- (b) the distances d_1 and d_2 are equivalent;

(c) the distances d_1 and d_2 are topologically equivalent

Exercise 0.8 Let $(V, +, \cdot)$ and let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V . Let V'_i be the dual space of $(V, \|\cdot\|_i)$ for $i = 1, 2$. Show if the norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent then $V'_1 = V'_2$.

Exercise 0.9 Let $(V, +, \cdot)$ be a vector space on the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$ and let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V . Show that if $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent, then

$$(V, \|\cdot\|_1) \text{ Banach} \quad \Leftrightarrow \quad (V, \|\cdot\|_2) \text{ Banach} \quad .$$

Hint: Assume that $(V, \|\cdot\|_1)$ Banach space. Let (v_n) be a Cauchy sequence in $(V, \|\cdot\|_2)$. Then write

$$C_1 \|v_n - v_m\|_1 \leq \|v_n - v_m\|_2 \leq C_2 \|v_n - v_m\|_1 \quad \forall n, m \in \mathbb{N}.$$

In the next lecture, we will introduce an important class of Banach spaces called Hilbert spaces. These are defined through the notion of scalar product defined on vector spaces.